# A Translation Floor for Cross-Lingual Archives: Spelling and Content Analysis in *Imvo Zabantsundu*

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## Abstract

**In its 1884–1894 issues, *Imvo Zabantsundu* published an isiXhosa section and a smaller English one. This study asks whether they differ in content once the language signal is removed from LaBSE embeddings, and whether modernising isiXhosa spelling meaningfully changes the answer. A classifier's accuracy on the newspaper's translations, the floor, estimates how well it tells the sections apart without much difference in content. Performance above this floor suggests that the sections differ in content, mainly in subject matter. Modernising the spelling brings translations slightly closer, but its effect on the content difference is inconclusive.**

**Keywords:** historical text normalisation; multilingual sentence embeddings; isiXhosa

## Introduction

*Imvo Zabantsundu*, founded by John Tengo Jabavu in 1884, published mainly in isiXhosa, with some English translations and other items (Schoots, Ngwendu, et al., 2023). NLP tools are mostly developed from modern text in high-resource languages (Joshi, Santy, et al., 2020), which gives researchers a reason to modernise the spelling of isiXhosa texts first, for example changing *Tixo* and *umteto* to *Thixo* and *umthetho*. Historical normalisation is judged by spelling accuracy or task scores (Bollmann, 2019), rarely by whether it changes a conclusion, making it a "forking path" (Gelman & Loken, 2013).

**Research question.** *After removing the language signal from a multilingual encoder's embeddings, can a classifier still distinguish between Imvo's two sections based on their content, and does modernising the spelling change the size of this difference?* **H1:** A classifier predicting which of the two sections a chunk (a 60-word piece of a page) comes from should achieve a higher balanced accuracy (BA; chance is 0.5) on all test chunks than on translated pairs. Pairs are chunks matched to their likely translation from the other section. Each pair should share the same content, so BA on these pairs is the floor, an estimate of how well the classifier tells the sections apart without a content difference. The difference between the two BAs, named $\Delta$, estimates the content difference. H1 is supported if the 95% confidence interval (CI) of $\Delta$, resampled over test issues, lies above zero. **H2:** Modernising isiXhosa spelling should lower the floor, since the encoder reads modern text better, while leaving $\Delta$ unchanged within $\pm$0.02 (measurement invariance). Beyond this range, modernising changes the conclusion.

### Background

**Historical isiXhosa.** The archive's machine-read (OCR) text was corrected by native isiXhosa speakers and deliberately not spell-checked, as 1880s spelling differs from today's (Schoots, Ngwendu, et al., 2023). Spelling is not neutral. Written isiXhosa was first shaped by missionary grammars made for learners rather than speakers (Maseko, 2017), and low-resourcedness is also a question of who builds tools and for whom (Nekoto, Marivate, et al., 2020).

**Removing language.** LaBSE (Feng, Yang, et al., 2022) was trained on translations in 109 languages, including isiXhosa, but its embeddings still identify the language. This signal can be removed by subtracting each language's average embedding (mean-centring; Libovický, Rosa, et al., 2020), which also removes average content differences, or by projecting out a few directions (Yang, Yang, et al., 2021; Ravfogel, Elazar, et al., 2020). As with control tasks (Hewitt & Liang, 2019), the newspaper's own translations are the control.

## Methods

**Data.** From the 517 Imvo issues in the archive files, 511 with at least 1,000 words per language are used. Chunks in the wrong language and text repeated across issues are left out, leaving 45,465 isiXhosa and 22,927 English chunks. Issues are split 357/77/77 (train/validation/test), stratified by year.

**Spelling modernisation.** A word list is built from 3.3M tokens of modern isiXhosa (mostly OPUS-100 (Zhang, Williams, et al., 2020), and also Wikipedia and Vukuzenzele (Lastrucci, Rajab, et al., 2023)). *Aspiration* restores the *h* that 1880s spelling left out after letters *t, p, k, r, c, q, x*, but only when the old spelling is almost absent from the modern word list ($\leq$2 uses) and the new one is common in it ($\geq$5). *Prefix* joins prefixes written apart (*e Palamente*$\rightarrow$*ePalamente*) only when the joined form is in the word list ($\geq$5 uses). The rules rewrite 9.1% of the 3.7M isiXhosa words, including 191 likely names (capitalised in $\geq$90% of uses). *Modern* spelling applies both rules, while *raw* is the original.

**Embeddings and translated pairs.** Each chunk is embedded with LaBSE and, for comparison, multilingual-E5 (Wang et al., 2024). Within each issue, an isiXhosa and an English chunk that are each other's closest match by cosine similarity create a pair if their similarity is $\geq$0.55 (622 of the 10,289 test chunks form 311 test pairs). Of 100 sampled test pairs, a fluent isiXhosa speaker labelled 24 herself and reviewed 27 that a team member had labelled with AI translations and suggested labels. Of these 51, 31% were translations, 29% partial and 39% different, suggesting that the 0.55 threshold is too loose. Her own labels find more pairs different (54% vs 26%), so the reviewed ones may lean towards the AI, though $\Delta$ on pairs labelled translation or partial is similar (0.24, 0.26).

**Removing the language signal (pair removal).** The differences within training pairs, which are mostly language, are stacked and their top $k$ directions are projected out of every embedding:

$$D = \left[x_i^{\text{xh}} - x_i^{\text{en}}\right]_{i \in P}, \qquad \tilde{x} = x - V_k^{\top} V_k x, \tag{1}$$

where $V_k$ holds the top-$k$ right singular vectors of $D$. A logistic regression, weighted so both sections count equally, is trained on training issues to predict each chunk's section. Two settings are chosen on validation issues: the number of directions removed, named $k$, is the smallest value in $\{0, 1, 2, 4, \ldots, 64\}$ that brings the floor to 0.55 or below (64 if none does, as here), and the regularisation strength $C \in \{10^{-2}, \ldots, 10^{3}\}$ is chosen for the highest BA. **Spelling effects.** For H2, each spelling

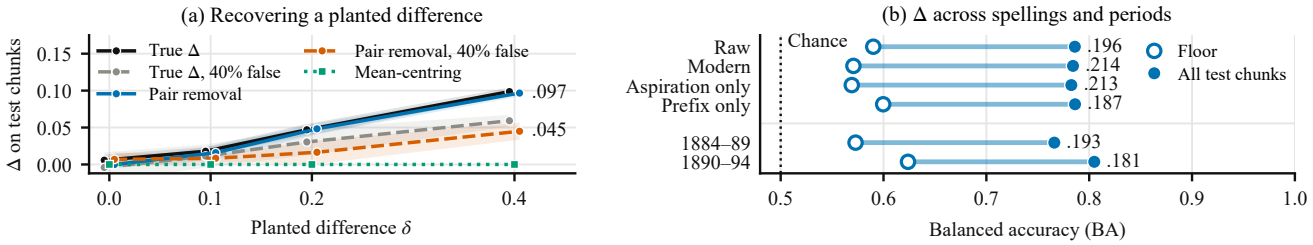



**Figure 1:** (a) Simulation, showing Δ is found as the planted difference δ grows (over 5 seeds with ±std). Dashed lines have 40% false pairs. Mean-centring subtracts each language's mean. (b) Test issues with LaBSE. Each line runs from the floor (open) to BA on all test chunks (filled), and its length, printed beside it, is Δ.

variant keeps the pairs and $k$ chosen from the original text, so only spelling is changed. The variant is equivalent to raw if the 90% CI of its change in Δ lies within ±0.02.

**Topic clustering.** Topics are $k$-means clusters of the language-removed embeddings, fitted without section labels. To test how much of Δ is subject matter alone, a *topic-only rule* is applied. It assigns a chunk to the isiXhosa section if its topic's isiXhosa share in training issues is above the overall share (as in Figure S1). This, and the check of where errors fall, were performed after the main results, so they are exploratory.

**Synthetic check (planted gap).** Each simulated chunk is a topic vector plus a language vector. A content difference is planted: English chunks use all topics evenly, while a share δ of isiXhosa chunks use only a quarter of topics. Simulated pairs copy isiXhosa content into English, and a classifier on topic vectors alone measures the true Δ. This checks the language removal and the floor, not LaBSE.

## Results

**The method recovers planted gaps.** Tested first on simulated data with a known answer (Figure 1a), the method recovers Δ (0.097 vs true 0.099 at $\delta{=}0.4$). With 40% false pairs (unrelated chunks recorded as translations), it shrinks (0.045 vs a true 0.059). But real false pairs may behave differently, as they are paired for looking similar.

**The classifier is above the floor (H1).** On Imvo, the classifier reaches 0.786 on all test chunks but only 0.590 on translated pairs, so Δ is 0.196 (0.167–0.224), supporting H1. Of 311 test pairs, it gives both halves one label for 201 pairs (it cannot tell which is the translation), correctly tells them apart for 83 and labels both the wrong way round for 27. Δ is positive in every Table S2 setting (smallest 0.046) and both time periods (Figure 1b), but its size depends on the pairs used (0.117 to 0.286 across the pair sets in Table S2). No $k$ reached the 0.55 target. Removing one direction lowered the test floor from 0.977 to 0.593, and removing more barely changed it. Removing language with only pairs of similarity ≥0.65 raised the floor to 0.725, as only 362 training pairs remain.

**Topic accounts for much of Δ.** A topic-only rule reaches 0.661 on test chunks, but only 0.523 on translated pairs, whose halves should share a topic. Topic accounts for about 70% of Δ (0.138 of 0.196; Table S2). Chunks on topics the other section usually covers are correctly predicted 59% of the time, against 88% for chunks on their own section's usual topics. Topic is not everything, though, as within one topic the classifier still reaches 0.73.

**Spelling's effect on Δ is unresolved (H2).** Aspiration cuts words missing from modern text from 31.5% to 24.5%, and modernising moves translated pairs closer (similarity +0.005). Yet its effect on Δ cannot be told apart from zero (+0.018, 90% CI −0.003 to +0.037; Figure 1b, Table S1; likewise for E5 encoder), and the floor's fall (0.019) is within noise. Every CI crosses ±0.02, so H2 is neither confirmed nor rejected.

## Discussion

**Subject matter is the clearest difference.** The sections are known to differ (Schoots, Ngwendu, et al., 2023), so what is new is measuring how much once language is removed, and showing that cleaning the spelling does not detectably change the answer. The English section leans towards colonial policy: commissions, land laws and liquor licensing; the isiXhosa section towards community life: association meetings, sport, marriage, land sales and market prices (Figure S1). The classifier errs where one section printed what the other usually did: 23 of 24 English land-sale notices go to the isiXhosa section, and most isiXhosa land-law and liquor articles go to the English one.

**Translation is an imperfect control.** The floor assumes that a translation says the same thing, yet the isiXhosa speaker noted the saying *isiXhosa asitolikwa*, "isiXhosa cannot be translated": even good translations adapt, turning "a feather in its cap" into the *indwe*, the crane feather of honour (Y. Siko, pers. comm.). If LaBSE misses meaning that may sometimes be attached to culture, the floor on translations may control content only roughly.

**Modernisation and the historical record.** Modernisation rules rewrite how people named themselves in 1884: a people (*Xosa*), a community (*Tembu*), a town (*Mtata*) and God (*Tixo*). The tools also read English more easily (2.11 vs 1.26 LaBSE tokens per word), which may tempt researchers to study only the English half (Nekoto, Marivate, et al., 2020). **Limitations.** Only 51 pairs were hand-checked and no one checked the rewrites. Topic checks use LaBSE only, and the method measures what was printed but cannot say why. Future archive studies should report the floor under both spellings.

## Acknowledgements

The authors thank Yonela Siko, a fluent isiXhosa speaker, for checking the 51 translated pairs used to verify the floor.

## Contribution Statements

We, Gopolang Mohlakola, Tawananyasha Kadango and Lily de Melo, worked together fairly on this project.

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13. **New Assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The code and normaliser are documented in README.txt, and the hand labels are in data/pair_labels.csv.

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14. **Crowdsourcing and Research with Human Subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [NA]

    Justification: No crowdsourcing or research with human subjects. A fluent isiXhosa speaker labelled 24 sampled translated pairs herself and reviewed 27 more that a team member had labelled with AI translations and suggested labels (translation, partial or different), following a written labelling guide. No personal data was collected.

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    Answer: [NA]

    Justification: No human subjects: the texts are newspapers from 1884–1894, and the archive's authors note that no ethics review was needed because the materials discuss no living subjects (Schoots et al., 2023).

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**Wits University Faculty of Science post-graduate student AI declaration**

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools ☐

I **did** make use of generative AI tools for the following (tick all that apply):

| | | |
|---|---|---|
| 1. | Idea Generation (research problem/design, hypothesis) | X |
| 2. | Sourcing Related Work (summarising, identifying sources) | X |
| 3. | Methods and Experiment Design (experiment setup, model tuning) | X |
| 4. | Data Analysis (presentation, coding, interpretation) | X |
| 5. | Theoretical Development (theorem proving, conceptual analysis) | |
| 6. | Code Development (generating algorithms, writing scripts) | X |
| 7. | Presentation (rendering graphics, formatting) | X |
| 8. | Editing (grammar, readability) | X |
| 9. | Writing (text generation, document structuring) | X |
| 10. | Citation Formatting (structuring, organising) | X |

If other uses were involved, please specify below:

| **Generative AI tool used (list all)** | **Used for?** |
|---|---|
| | |
| | |
| | |
| | |

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

**Student number**:

2545080 2680747 2588206

**Candidate signature**:





**Date**: 30/10/2026

## Supplementary material

Two tables and a figure that support claims in the body, from the saved results in `results/`.

**Table S1:** Spelling variants on test issues with LaBSE. The last column is each variant's change in $\Delta$ from original (raw) spelling, with its 90% CI (1,000 resamples of test issues).

| Variant | All | Floor | $\Delta$ | $\Delta - \Delta_{\text{raw}}$ (90% CI) |
|---|---|---|---|---|
| raw | .786 | .590 | .196 | – |
| modern | .784 | .571 | .214 | $+$.018 ($-$.003, $+$.037) |
| aspiration only | .783 | .569 | .213 | $+$.017 ($-$.001, $+$.038) |
| prefix only | .786 | .600 | .187 | $-$.009 ($-$.024, $+$.006) |

**Table S2:** $\Delta$ across checks, on test issues with LaBSE and original (raw) spelling unless stated. Brackets give the 95% CI where available (1,000 resamples of test issues). The top five rows use the same classifier and differ only in which pairs give the floor, and the two in bold set the range of $\Delta$ in Results. Pairs matched on shared names and numbers are found without embeddings. The topic-only rule reruns the clustering in Figure S1, so cluster sizes differ slightly. kNN is a cosine $k$-nearest-neighbour classifier.

| Setting | All | Floor | $\Delta$ (95% CI) |
|---|---|---|---|
| **Translated pairs hand-checked as full translations (16 pairs)** | **.786** | **.500** | **.286 (.142, .412)** |
| **Pairs matched on shared names and numbers, but chunks not in translated pairs (252 chunks)** | **.786** | **.670** | **.117 (.066, .169)** |
| All translated pairs, similarity $\geq$0.55 (311 pairs) | .786 | .590 | .196 (.167, .224) |
| Pairs matched on shared names and numbers (217 pairs) | .786 | .627 | .159 (.128, .191) |
| Translated pairs with similarity $\geq$0.60 (161 pairs) | .786 | .547 | .239 (.203, .276) |
| Topic-only rule | .661 | .523 | .138 |
| E5 encoder | .877 | .618 | .258 (.216, .304) |
| kNN classifier, E5 encoder | .599 | .553 | .046 |

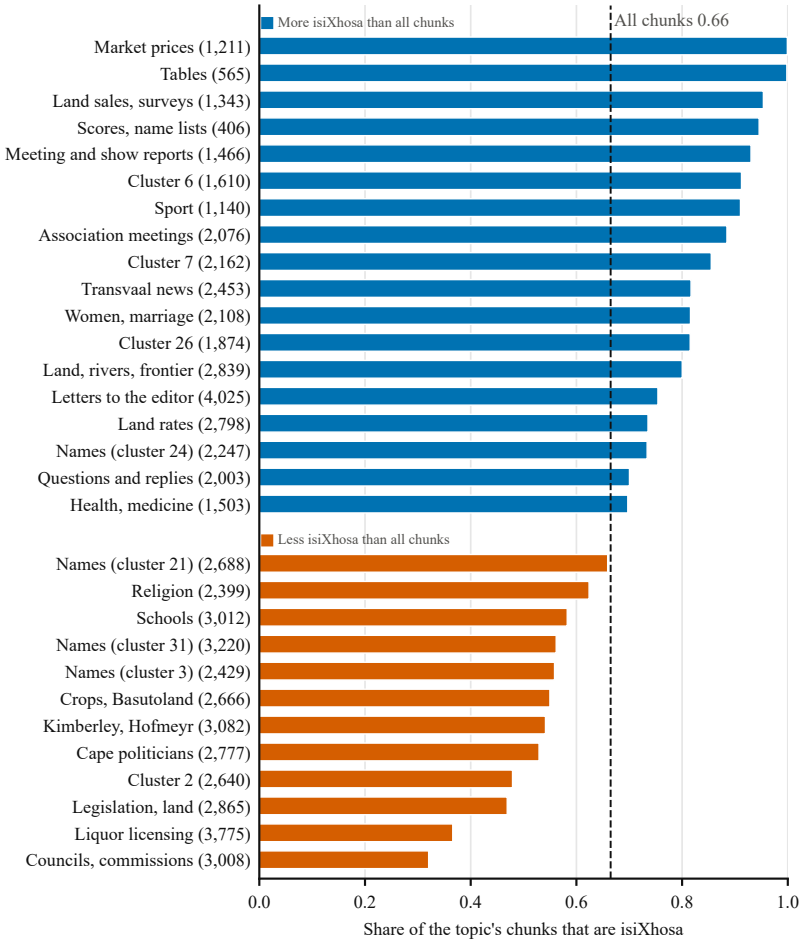



**Figure S1:** What each section usually wrote about. Each bar is one of 30 topics, with its chunk count, and shows the share of its chunks that are isiXhosa. Blue topics are above the overall share (0.66) and orange ones are below it. Topics are $k$-means clusters of the language-removed embeddings, named by the authors from their most distinctive words. Clusters labelled *names* hold mostly personal and place names, and numbered ones have no clear subject. The topics overlap (silhouette score 0.008, where 0 means fully overlapping), so they describe content rather than measure it.